



Provisional Crowns

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In this lecture we will discuss the fabrication of a direct provisional crowns.

Provisional Crowns



Provisional crowns or temporaries are a common procedure for indirect restorations. A provisional serves a few purposes, they protect the exposed tooth structure, protect the gingival tissues and maintain position of the surrounding teeth. The last one here is very important because you want the permanent restoration to fit when it comes back from the lab. Dr. Varimezova will go into more detail about the materials and some different ways that a temporary can be fabricated. We will focus on the direct methods for provisional fabrication.

Bisacrylic Materials



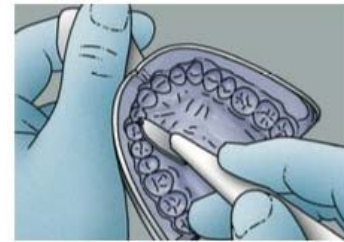
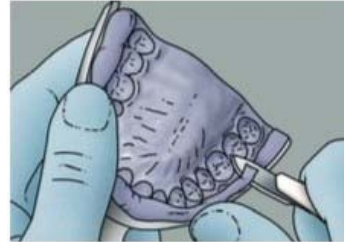
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The most common material used today in practice for a temporary crown is bisacrylic materials. Most manufactures make these products. We will be using: Paradigm Temporary Material from 3M. These are binary materials that need to be mixed. These materials are not mixed in a 1:1 ratio. As you can see one side of the cartridge is larger. You will need a specific mixing gun to mix up these materials.

Making a Provisional



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The first step to making a provisional crown is to have a matrix to work with. This can be a pre-impression of the area if the tooth is fully intact or duplication of a wax-up. Materials such as Alginate, which is a form of Polyvinyl siloxane (PVS), are commonly used to take the pre-impression. Clinically, you can use a triple tray and impress the arch that you are working on. In the Sim Lab, we will have you impress both the upper and the lower so that one impression can be used for both arches.

To make this impression, the Alginate is placed into the gun with a mixing tip. You will want to overload the tray so that the material will extend beyond the gingival margin of the tooth. A helpful trick when doing this on a patient is to massage the cheek or lip to get better adaptation. This is not possible with our mannequins. The material will have a specific setting time, once that has been fulfilled take out the impression and evaluate it.

When looking into the impression you want to make sure that the bite looks good. It will have the appearance of light passing through the material. Next, you will want to make sure that there are not any large bubbles in the impression or voids. You will trim unwanted excess around the tray and cut out the interproximal septa between the teeth if any. Lastly, you will want to try the impression back in to make sure it fits properly.

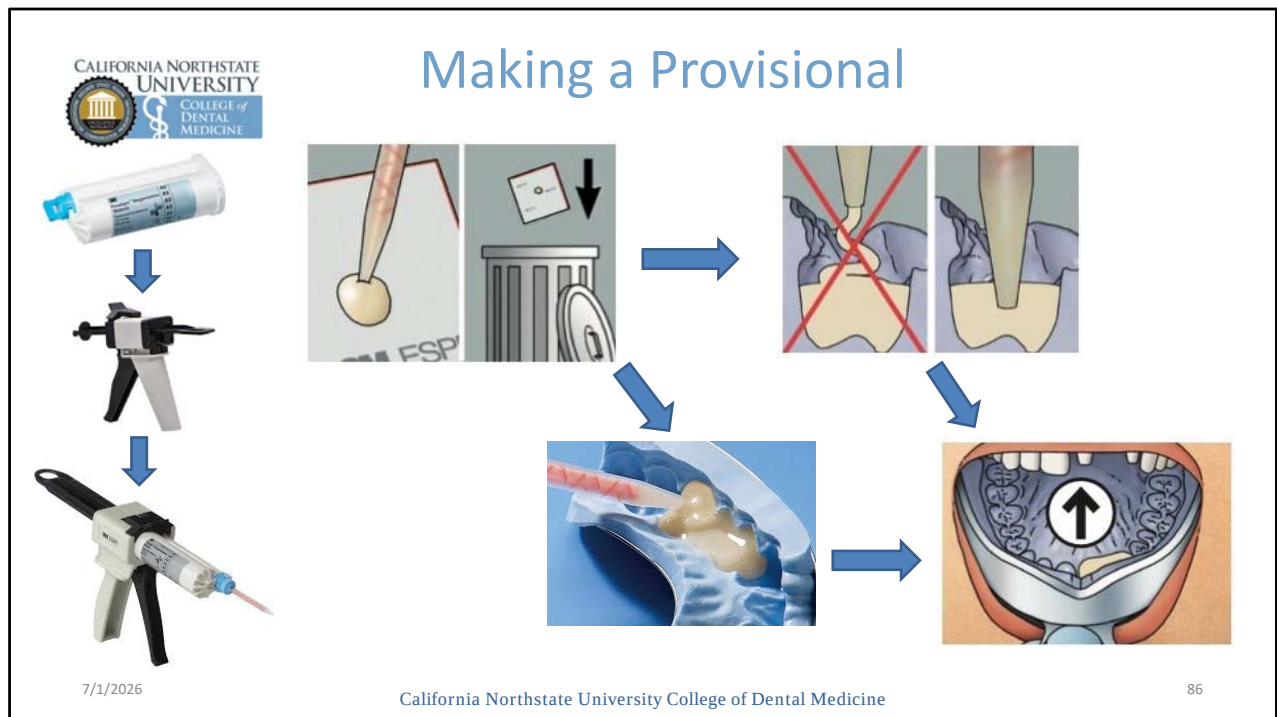
Another technique is to use a PVS putty. This will come in a base and catalyst that will need

to be mixed in your hands. The putty is then placed onto the teeth and molded around the area that you will be working on. It is advised to make sure the putty covers the tooth you are working on as well as at least the tooth mesial and the tooth distal of the tooth to be prepared for a crown. The putty is then allowed to set and then is removed and trimmed as necessary.

Making a Provisional



Now that you have some for of stent or impression you are ready to make a provisional. Following the preparation of a tooth, you will want to try in your stent or impression to ensure proper fit and that you know how to insert it properly. In the simulation lab you will want to place petroleum jelly on the prepped tooth and neighboring teeth. This will prevent the material from sticking to the tooth structure. Clinically, this is not necessary due to saliva and the material itself.



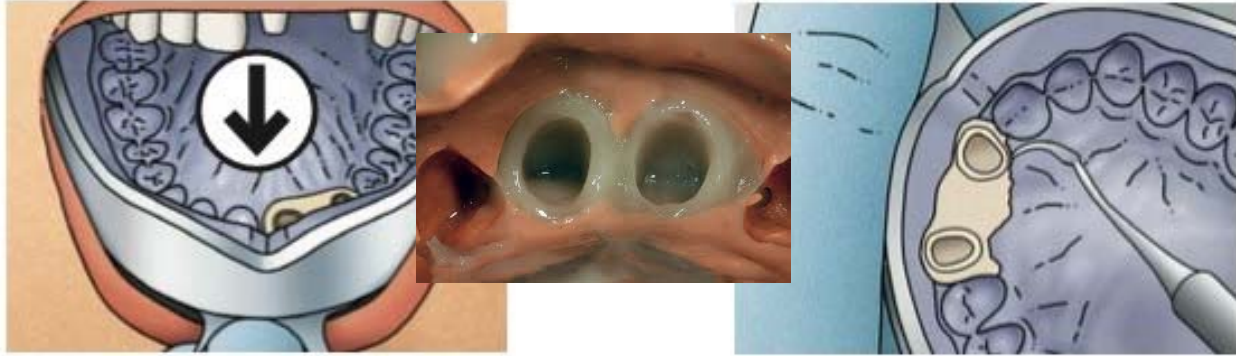
The first step is load the temp material into the proper dispensing gun and place the appropriate mixing tip onto the cartridge. When you are ready to start, express a little of the material onto your glove or a mixing pad. This material will be thrown away. You generally do not want to use the initial amount whether it is a new cartridge or using an old one. The initial amount can be contaminated in a used cartridge or have an oily consistency on a new cartridge.

Place the tip all of the way into the negative of the tooth to be temporized and the material should be expressed filling the negative of the tooth. The impression or stent is then placed into the mouth and the patient is asked to bite down to hold it place.

Video of Loading Temp Material



Making a Provisional



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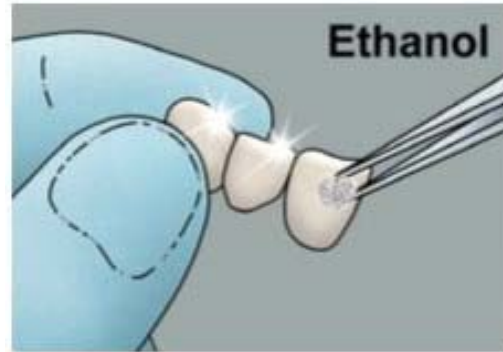
For the Paradigm material you will leave it in the patient's mouth for 1 min and then have the patient open and remove the impression or stent. Most of the time the temporary will come out in the impression or stent. Leave it there undisturbed until it is set. If you try and alter the temp you could distort the temporary while it is setting. It will take 4 min to set up out of the mouth.

Making a Provisional



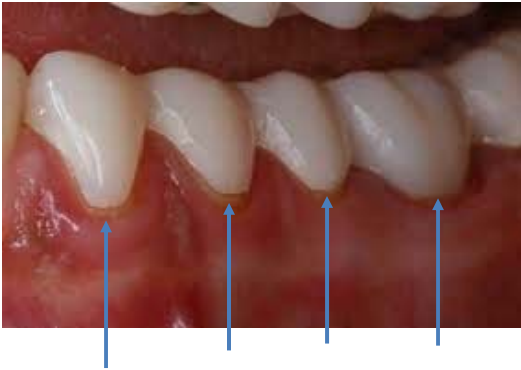
Once the temp material has set it is time to trim the excess. This can be done with an acrylic bur, wheels and points or with a highspeed and finishing burs. Remove any excess up to your margin. Be careful not to remove too much. It is better to leave a little excess at the margin and try it on the tooth than to remove too much and be required to make a new provisional. Your loupes will certainly be helpful in visualizing the negative of your margin in the provisional. To identify your margin in the provisional, you will want to look for an abrupt change in direction. This will indicate where the margin ends and excess material starts. This is indicated with an arrow in the left side.

Oxygen Inhibited Layer



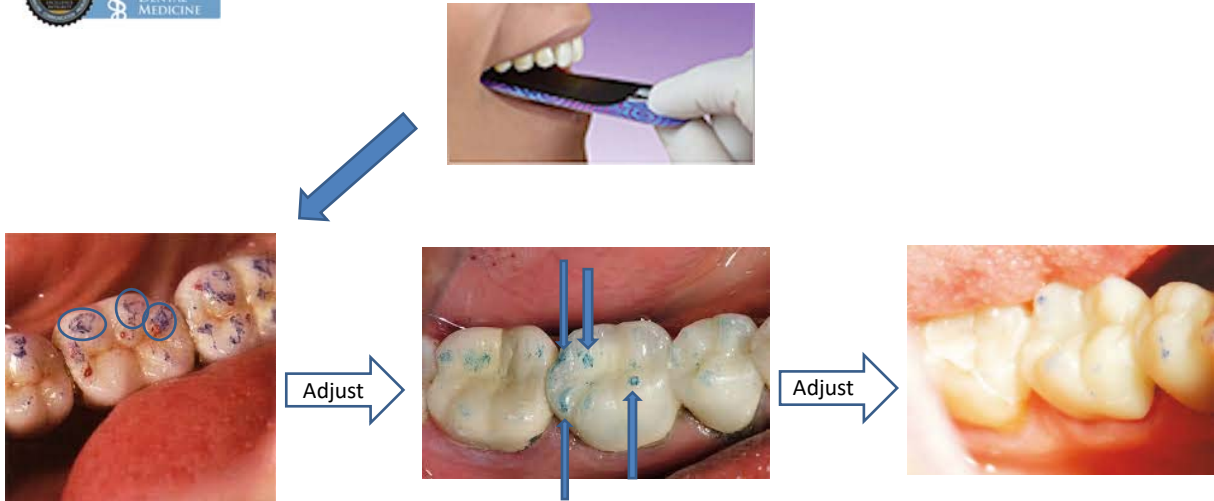
Like composite restorations, the bisacrylic material has an oxygen inhibited layer on the surface. This allows you to add composite to the surface to fill voids, close an open margin or change the shape of the temporary to mimic the neighboring teeth. In general, flowable composite is probably the best to use to fill voids and gaps. When a large changes need to be made, composite paste will need to be used. If you do not need to add composite, you can clean the surface of the provisional with ethanol which will remove the oxygen inhibited layer.

Temporary Evaluation



As with the seating of a permanent crown, it is important to check the margins of the crown as well as the interproximal contact. The margins of a provisional will not be as exact as the final crown but you want the margin sealed without large amounts of extra material. Too much temporary material beyond your margin can cause gingival irritation and bleeding which can complicate the seating of the final restoration. Next, check the interproximal contact with floss. It is important to hold the crown down when you are flossing. Floss down and then pull out to the side. If the contact is too heavy adjust as necessary. If it is too light, add to the contact area with composite paste until you are happy with the contact.

Occlusion



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Once you are happy with the temporary, it is time to check the occlusion. Using articulating paper have the patient bite down a few times. Check the provisional. If the occlusal marks are heavy, adjust and repeat as necessary until a light occlusion is obtained. If a heavy occlusion is left, a patient can experience post operative sensitivity, pain and even swelling.

Preparation Check



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In the posterior crown lecture, we should a few ways to check the occlusal reduction. The temporary crown is another way to check your occlusal reduction. The process to do this is to make your temporary before you impress or scan and then use your calipers to measure the thickness of the temporary crown on the occlusal surface. If it is not the measurement you want, you can then go back to the tooth and reduce more to gain the ideal reduction.

Polishing



The paradigm material will have a gloss to it so there is generally no need to polish the provisional. Now if you need to adjust areas other than the margin, polishing may be needed. This can be done with composite polishers to gain a high shine.

Seating

BAD



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Good



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The seating of a provisional is very similar to a final crown. You will want to check the contacts, the margin and the occlusion. Adjust the contacts, margin and occlusion, as necessary. Unlike a final crown, you will want to limit the amount of temporary cement used. Dispense the temporary cement onto a mixing pad and mix the base and catalyst. Then take a small brush and brush the temporary cement on the margin. Seat the provisional on the tooth and have the patient gently bite down on a cotton roll. In a minute or two the temporary cement will set up and is removed with an explorer and floss. As with a final crown you will want to tie a knot into your floss and run it back and forth through the embrasure to remove any residual cement. When using this technique generally there will be very little excess temporary cement.

Final Steps



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Once the temporary has been seating and the temporary cement has set it is time to clean the excess. This is done with an explorer and floss with a knot in it. Lastly, you will want to sit your patient up into a normal seated position and have the patient check their bite again to verify their bite is comfortable.